Intelligent System for the Learning of Sign Language Based on Artificial Neural Networks

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Abstract. Sign language has become a form of communication for people who have some type of hearing impairment, in order to relate to the world around him and perform daily activities including education, work performance and personal development. In the knowledge society, the development of new Technologies for Information and Communication (TIC), they constitute a teaching tool that changes the conditions for learning and communication as the world is digitized. Therefore, the development of an Ecuadorian sign language learning system based on the use of a gesture sensor and an artificial neural network multilayer feed-forward, implementing the Backpropagation algorithm, which takes advantage of the parallel property of decreasing the time required by a processor to distinguish the existing relationship between given patterns, represents a solution for improving the teaching-learning process.

The system consists of three main layers: The first is responsible for the acquisition of data through a gestural device; the second uses the library of the gesture sensor to perform the acquisition and storage of information of the hand and the position of the fingers. Finally, the output layer, once the training of the neuron is analyzed, generates the real-time recognition of sign language.

Keywords: CSCL · Artificial neural networks · Pattern recognition · Leap Motion

1 Introduction

The World Health Organization (WHO) reports that more than 5% of the world population suffers from some type of hearing disability, among them, 466 million are adults and 34 million are children [1]. In Latin America, the presence of hearing problems is 1.6% in children under 14 years of age; in people older than 15 years it is 8%; and 38.62% corresponds to people over 65 years of age. In Ecuador, according to the National Council for Disability Equality (CONADIS) there are 437,268 people with some degree of disability; Of these, 14.17% have hearing impairment, subdividing this percentage of the population as follows: 8.13% are children under 12 years, 13.13%

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people between 13 and 65 years, and 19.05% people over 65 years. Of a total of 58,161 people with occupationally active disabilities, 16.03% are people with hearing disabilities, of this percentage 16.24% are people whose age is between 18 to 29 years, 78.49% people between 30 to 64 years and 5.28% they are people over 65 years [2].

Traditional education is mostly focused on people without any type of disability, which makes it difficult for people with disabilities to have access to a quality education that has the improvements and technological advances developed throughout the world [3]. Therefore, promoting the inclusion of people with disabilities and developing new teaching tools is considered an appropriate option in the field of education, new methodologies such as Computer-Supported Collaborative Learning (CSCL) are presented as an emerging area of the learning sciences regarding how people can interact together with the help of computers [4].

There is an interest in projects dealing with artificial neural networks, or gesture sensors, focused on the teaching of sign language, as can be indicated in the following works; for example, in the article developed by [5], consists of the use of Leap Motion, a computerized pattern classification module and a graphical user interface for the teaching of Ecuadorian sign language.

The proposal made by [6], consists of a Sign Language Teaching Model (SLTM) designed to develop different Communication Skills (CS) in deaf children within a Collaborative Learning Environment with Mixed-Reality (CLEMR). In the work developed in [7], he proposes the use of a new information technology to model sign language based on the human computer interaction model. In addition, in [8] the recognition of patterns through artificial neural networks is proposed.

Based on the above, a learning system is presented that uses as input device a gesture sensor, which acquires the complete data of the hand such as: size of the fingers, center of the palm, orientation with respect to the axes; performs the recognition and classification of the signs through the development of a neural network [9], on the other hand, it contains three environments created especially to improve the learning experience of sign language, to contribute in the process of educational and professional training of people with some degree of hearing disability.

This work is divided into six sections, including the introduction. The second section deals with the training of the artificial neural network, in Sect. 3 the development of the system interface is exposed; Sect. 4 contemplates the tests performed on the system on users with and without disabilities, the presentation of the results analysis is found in Sect. 5 and finally the conclusions are located in Sect. 6.

2 Neural Network Classifier

In the recognition phase, a multilayer feed-forward Neural Network is used, with the Backpropagation algorithm; which takes advantage of the property of reduction of time required by a computer processing unit; to distinguish the existing relationship between given patterns [10]. The neural network has three layers: input, hidden and output; the first is constituted by m entries, the hidden layer is composed of n neurons and a logarithmic sigmoid activation function f^1, the output layer is composed of six neurons with a logarithmic sigmoid activation function f^2; this structure can be seen in the Fig. 1.

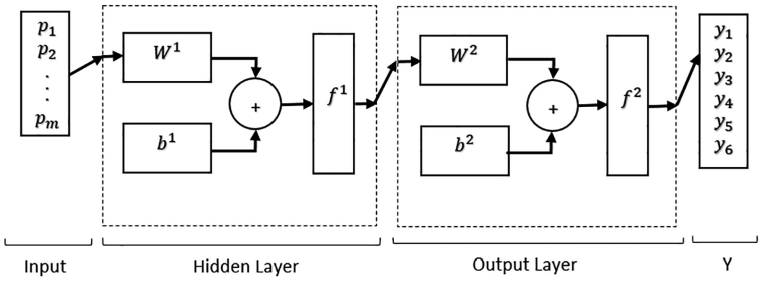


Fig. 1. Architecture of a three-layer artificial neural network feed-forward.

The output given by the supervisor is responsible for establishing the response that forms the network from a specific input, in this context we have the binary combination 000001 which is equal to 1, up to 100111 equivalents to 39; the same that represents the letters A to Z and to the numbers from 1 to 9 respectively.

The vector Y of the Neural Network, for an input vector P = (P1; …; Pm), is given by the following expression:

$$Y = f^2\left[W^2 f^1\left(W^1 P + b^1\right) + b^2\right] \tag{1}$$

When the output Y, express a probability that P belongs to a letter of the alphabet, W1 and W2; they express the matrices of the weights of the first and second layers respectively, b1 and b2 determine the level of threshold or activation of the neuron of the input and hidden layer.

The activation function used in the Artificial Neural Network is a logarithmic sigmoid function because the output vector has values of 0 and 1, f1 = f2 = 1 = $(1 + e^{(-z)})$.

To perform the training of the neuron, a database was created in Amazon Web Server (AWS) with values of the angle between the normal vector and the direction of the palm of the hand, angles of the palm with respect to the x, y, z; the angles generated by the direction vector of the following phalanx, except the distal phalanx in the thumb. If the vector of entries belongs to a letter, the output will give us an approximate vector, for example, if letter is A, the output gives us a vector equal to [0 0 0 0 0 1], from number 9 the output is [100111].

3 Development of the Neural Network

The implemented network is a multilayer perceptron, structured by an input layer that has a vector with 21 features, a hidden layer with 25 neurons and an output layer with 6 neurons. Weights and tracks were initialized with randomic values between −0.5 and 0.5. The output layer and the hidden layer have a logarithmic sigmoidal activation function that limits the value of the output between 0 and 1, adopting values very close to 0 when the input is small, but as the value increases on the abscissa axis the function happens to be growing.

The weights and the values of the roads were calculated by the method of back propagation, which starts from the last layer to the first, which causes the increase in error; to find the gradient of the error totalized in each neuron, the new weights and paths are recalculated to the inverse direction together with the value of Alpha, which for this network has a value of 0.5, which represents the normal learning rate avoiding jumps in said learning, Fig. 2.

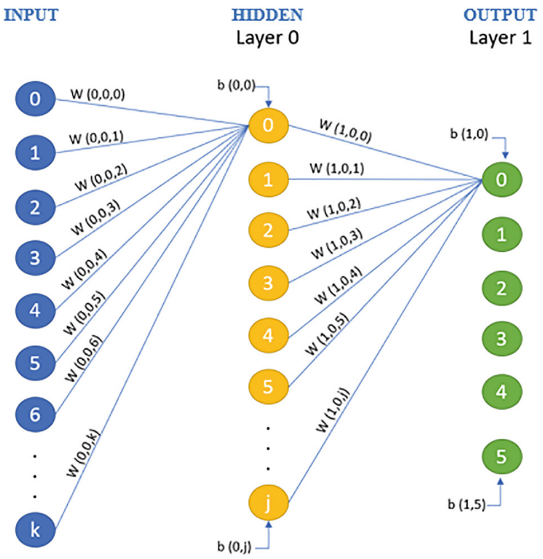


Fig. 2. Graphic description of the developed neuronal network

4 Structure of the System

The architecture of the system consists of three main layers: Inputs, Script and Outputs Fig. 3. The input layer is responsible for the acquisition of data by means of a gestural device; the script layer uses the library of Leap Motion, to perform the acquisition of the position of the fingers when forming a signal, the storage of the information of the hand is done in MySQL, for the effect the Amazon Web Server platform also communicates the Matlab and MySQL programing interfaces for data processing and neuron training. Finally, the output layer generates the recognition in real time of Sign Language, by creating a teaching-learning system made in Unity 3D.

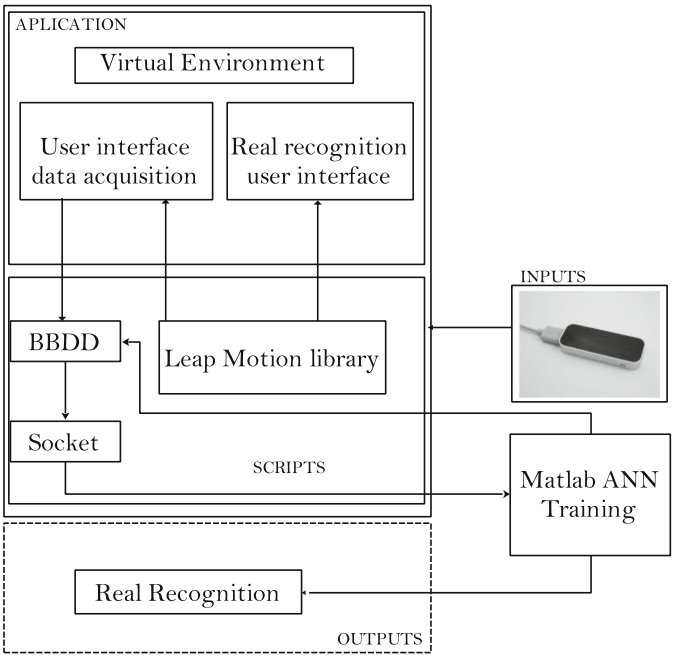


Fig. 3. Structural diagram of the system.

4.1 Description of the Graphical User Interface (GUI)

The Fig. 4a, presents the window of the environment menu with the following options: "Instrucciones", pressing this button gives access to a manual of necessary instructions for the manipulation of the system, "Aprendizaje", access directly to a window where the created learning environments are located, "Salir" closes the whole application.



Fig. 4. (a) Selection menu window. (b) Learning environment window

The Fig. 4b, presents the learning environment window, which consists of three environments: ordered letters, disordered letters and letter leakage.

The environment ordered letters, allows a game that offers the recognition of letters ascending form the A–Z, like numbers 1–9; this environment serves for user training and learning, there is no type of limit of use, nor time; if the user performs the indicated signal the learning system continues until analyzing the configurations and numbers presented Fig. 5.



Fig. 5. Ordered letters interface

The disordered letters environment, allows access to a new window where a set of 10 letters is presented in different order, in this environment the user can corroborate the level of learning.

5 Tests Performed

In this section the teaching-learning capacity of the sign recognition system is evaluated, which will be based on three aspects: response of the neural network recognition, learning tests and finally the usability of the system.

5.1 Response of the Neural Network Recognition

To verify the performance of the neural network, the analysis of the area under the curve (AUC) is used, using the receiver operating characteristic curves (ROC), which is a graphic representation between the success rate (probability of recognizing the sign done) against the false alarm rate (probability of not recognizing a signal) for tasks that have only two possible outcomes (yes/no, acknowledges/does not recognize).

Figure 6 shows the results obtained from the Total AUC of the static configurations, obtaining a maximum value of 0.9779, and of the dynamic configurations with a maximum value of 0.9700, therefore, it is considered as a system perfect classifier since it exceeds 0.5.

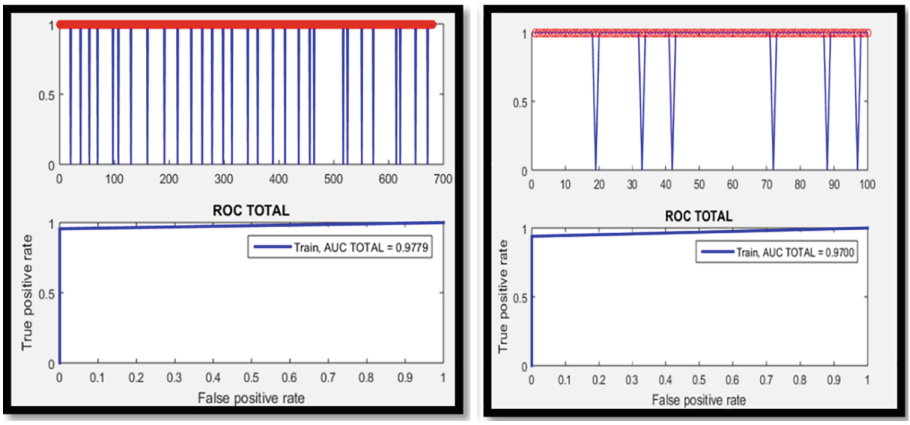


Fig. 6. (a) AUC - Static configuration, (b) AUC - Dynamic configuration

The Table 1 shows the results obtained from the area under the ROC curve, obtained from each static and dynamic configuration that forms the Sign Language.

Table 1. Area under the curve (static and dynamic configurations)

Nº.	Letters	AUC	Nº.	Letters	AUC	Nº.	Letters	AUC
1	A	0,9750	11	J	0,9750	21	R	0,9742
2	B	0,9742	12	K	0,9750	22	RR	0,9750
3	C	0,9742	13	L	0,9727	23	S	0,9742
4	CH	0,9750	14	LL	0,9750	24	T	0,9742
5	D	0,9750	15	M	0,9750	25	U	0,9750
6	E	0,9742	16	N	0,9742	26	V	0,9742
7	F	0,9712	17	Ñ	0,9750	27	W	0,9735
8	G	0,9750	18	O	0,9735	28	X	0,9750
9	H	0,9992	19	P	0,9750	29	Y	0,9992
10	I	0,9750	20	Q	0,9985	30	Z	0,9437

5.2 Learning Tests

The tests of the learning system are carried out with a total of 10 users, of which 4 are children who have hearing impairment and 6 children without hearing impairment.

The learning tests starts with a stage of recognition of the system by the user, then the three learning modes must be completed: Sorted letters mode, disordered letters mode and learning mode or letters leakage; in the first, the whole alphabet of the sign language is ordered in an orderly manner AZ, the second mode shows 10 letters of the alphabet interchangeably, and the third consists in making the greatest number of letters of a word in a time of 30 s at the end, the time is analyzed and the score obtained is indicated.

The learning curve on the part of the users who have auditory disability is appreciated, it can be observed that the performance and the level of learning increases in the 4th and 5th tests. The performance and level of learning of users who do not present any type of hearing impairment where it is observed that the learning curve increases exponentially.

Figure 7, shows the average learning curve obtained in children with hearing impairment versus children without hearing impairment, graphically you can see the similarity of the curves, which show an exponential growth, which allows to deduce that the system fulfilled the teaching objective.

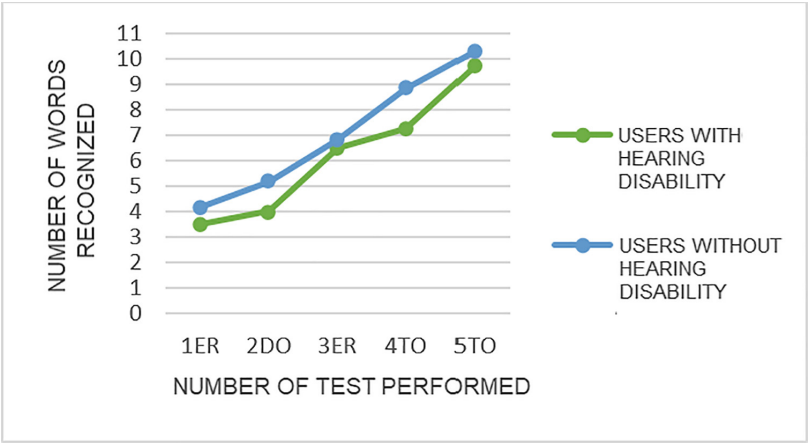


Fig. 7. Average learning curve of users with and without hearing impairment

6 System Usability

The SUS questionnaire allows evaluating the usability of the system once the learning is analyzed, a test is performed, collecting the data for analysis and study; The results obtained by the users are presented to each question posed [11].

For the system to be acceptable in terms of usability, the global average must exceed 68% according to the SUS questionnaire standard.

The results obtained in the questionnaire exceed the standard value by 80.25%. Therefore, it can be considered that the learning system developed is comfortable in its use, it does not present complexity in the manipulation and, above all, the results indicate that It is useful for learning sign language.

7 Analysis of Results

The system based on artificial neural network presented a better response in comparison to the statistical classifier, since it was obtained more than 97% of correct answers compared to 87%, derived by the use of the statistical classifier.

The multifaceted neural network was adapted correctly to the data entered by the user, allowing to maintain an interaction between the student and the system, which is evidenced by an improvement in learning.

The use of assistance systems for teaching in cases of people with special abilities, allows the student to progress in their learning as it does not depend on the presence of an instructor to practice the different signs.

The teacher can integrate computer tools in their teaching process, including people with disabilities, allowing them to contribute to the development of their daily activities.

8 Conclusions

The development of the Ecuadorian sign language learning system through neural networks is a tool that has allowed the inclusion of new technologies in the learning of basic sign language for children with and without deterioration of the auditory system.

The responses of the average learning curves performed in the 10 users with and without auditory disability show an exponential growth, which indicates that the higher the number of training sessions the learning level increases.

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